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Attentional blink suppresses both stimulus-driven and representation-driven cross-modal spread of attention

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Abstract

Previous studies have shown that visual attention effect can spread to the task-irrelevant auditory modality automatically through either the stimulus-driven binding process or the representation-driven priming process. Using an attentional blink paradigm, the present study investigated whether the long-latency stimulus-driven and representation-driven cross-modal spread of attention would be inhibited or facilitated when the attentional resources operating at the post-perceptual stage of processing are inadequate, whereas ensuring all visual stimuli were spatially attended and the representations of visual target object categories were activated, which were previously thought to be the only endogenous prerequisites for triggering cross-modal spread of attention. The results demonstrated that both types of attentional spreading were completely suppressed during the attentional blink interval but were highly prominent outside the attentional blink interval, with the stimulus-driven process being independent of, whereas the representation-driven process being dependent on, audiovisual semantic congruency. These findings provide the first evidence that the occurrences of both stimulus-driven and representation-driven spread of attention are contingent on the amount of post-perceptual attentional resources responsible for the late consolidation processing of visual stimuli, whereas the early detection of visual stimuli and the top-down activation of the visual representations are not the sole endogenous prerequisites for triggering any types of cross-modal attentional spreading.

KEYWORDS

attentional blink, cross-modal spread of attention, ERPs, representation-driven, stimulus-driven

1 | INTRODUCTION

In the research field of visual attention, it is well-documented that attending to one visual feature (e.g., color) in an object results in enhanced processing of not only the attended feature, but also other unattended visual features (e.g., shape) belonging to the same object, indicating visual attention can spread to task-irrelevant visual features within an object (Desimone & Duncan, 1995; Egly et al., 1994; Martínez et al., 2006, 2007; Melcher & Vidnyánszky, 2006; O'Craven et al., 1999; Schoenfeld et al., 2003; Wylie et al., 2004, 2006). Moreover, a series of event-related potential (ERP) studies have shown that the effect of visual attention can spread cross-modally to the task-irrelevant auditory modality automatically, either through the *stimulus-driven* binding process (Busse et al., 2005; Donohue et al., 2011; Fiebelkorn, Foxe, Schwartz, et al., 2010; Talsma et al., 2007; Zimmer et al., 2010) or via the *representation-driven* priming process (Fiebelkorn, Foxe, & Molholm, 2010; Molholm et al., 2007). Both the stimulus-driven and representation-driven attentional spreading effects were quantified by the occurrence of the attention-related negative difference (Nd) component

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(e.g., Fiebelkorn, Foxe, & Molholm, 2010; Fiebelkorn, Foxe, Schwartz, et al., 2010; Molholm et al., 2007; Talsma et al., 2007). The Nd is a prolonged ERP difference component manifested as a larger negative-going amplitude over fronto-central scalp region that begins approximately 150 ms after stimulus onset in response to attended than unattended auditory stimuli (Hansen & Hillyard, 1980), which is generally thought to reflect an enhancement of auditory processing due to selective attention (Giard et al., 2000; Näätänen, 1982; Singhal & Fowler, 2004). In terms of the inducing condition and underlying mechanism, the stimulus-driven spread of attention occurs whenever a task-irrelevant auditory stimulus is presented simultaneously with a spatially attended visual stimulus, thus, reflects a bottom-up, inherent audiovisual feature binding tendency based on the co-occurrence of visual and auditory stimuli (Busse et al., 2005; Fiebelkorn, Foxe, & Molholm, 2010; Fiebelkorn, Foxe, Schwartz, et al., 2010; Talsma et al., 2010; Tang et al., 2016). In contrast, the representation-driven spread of attention occurs only when the representation of a task-irrelevant sound (e.g., a bark) is congruent with that of the to-be-attended visual object category (e.g., an image of a dog), hence, reflects a top-down, high-level cross-modal priming effect based on the activation of highly learned associations between features in different modalities (Fiebelkorn, Foxe, & Molholm, 2010; Molholm et al., 2007; Talsma et al., 2010; Tang et al., 2016).

Accordingly, the extant literature seems to suggest that the spatial-based and object-based visual selective attention operating at the *perceptual* stage of processing are the only ***endogenous*** prerequisites for triggering both types of cross-modal spread of attention. However, it is well-established that the mechanisms of attentional selection are multifaceted, which can operate not merely at the perceptual stage but also at the *post-perceptual* stage of processing (Duncan, 1980; Duncan & Humphreys, 1989; Mangun & Hillyard, 1990; Vogel et al., 1998). Furthermore, the neural correlates of both types of cross-modal spread of attention (i.e., the Nds) were found to also emerge relatively late in auditory processing stream, beginning approximately 200 ms poststimulus and sustaining for another 200 ms or even more (Busse et al., 2005; Donohue et al., 2011; Fiebelkorn, Foxe, & Molholm, 2010; Fiebelkorn, Foxe, Schwartz, et al., 2010; Molholm et al., 2007; Talsma et al., 2007; Zimmer et al., 2010). Based on the two aspects of prior knowledge, it is reasonable to infer that adequate attentional resources operating at the late, post-perceptual stage of processing might be another endogenous prerequisite for the occurrences of both types of cross-modal spread of attention. Nevertheless, since the post-perceptual attentional resources have not yet been manipulated in previous studies of cross-modal spread of attention, it is currently unclear whether the stimulus-driven and representation-driven attentional spreading would be inhibited by the lack of post-perceptual attentional resources whereas the visual stimuli are still spatially attended and the visual target object categories still receive object-based attention (i.e., the representations of visual targets are still activated in a top-down manner). The answers to these questions would shed novel light on the boundary conditions, and hence, the underlying cognitive mechanisms, of both types of cross-modal attentional spreading.

The current study aimed at resolving these questions via ERP recordings and taking advantage of the ***attentional blink*** paradigm under which observers often fail to discriminate the second (T2) of two successive targets embedded in a rapid stream of visual stimuli that are all presented at the spatially attended location when the T2 is temporally close to the first (T1) (Raymond et al., 1992). Because the attentional blink phenomenon is well known to reflect a post-perceptual attentional mechanism that is most likely to suppress processing at the stage of working memory (Chun & Potter, 1995; Dux & Marois, 2009; Dux et al., 2014; Kranczioch et al., 2007; Olivers et al., 2007; Ptito et al., 2008; Raffone et al., 2014; Sergent et al., 2005; Vogel et al., 1998; Zivony & Lamy, 2016), it is the best candidate for manipulating the post-perceptual attentional resources in an orthogonal manner with respect to the spatial-based and object-based visual voluntary attention. Moreover, as many prior studies have shown that the attentional blink effect (i.e., lower T2 accuracy at short than long T1-to-T2 lag) was robust even when a salient sound was presented synchronously with the visual T2 (Adam & Noppeney, 2014; Haroush et al., 2011; Kranczioch & Thorne, 2013, 2015; Olivers & Van der Burg, 2008; Zhao et al., 2020), the attentional blink paradigm is suitable for manipulating attentional resources under the experimental design of cross-modal attentional spreading. Specifically, the present study manipulated object-based selective attention to the visual T2 (e.g., discriminating which dog the T2 was whereas no discrimination was required when the T2 was an image of either a car or a drum) under the attentional blink paradigm. Therefore, the current task design could investigate the stimulus-driven and representation-driven attentional spreading under different levels of ongoing post-perceptual attentional resources (i.e., short vs. long T1-to-T2 lag) even when all of the visual T2s were still spatially attended (for the stimulus-driven) and visual T2s of target object categories still received further object-based attention (for the representation-driven).

Nonetheless, there is a reason to make a reversed prediction regarding the effect of visual attentional blink on both types of cross-modal spread of attention: it has been reported that auditory processing of task-irrelevant sounds was enhanced, instead of suppressed, during the visual attentional blink, which was interpreted as a reduction in active suppression of the irrelevant auditory processing (Haroush et al., 2011). Therefore, it seems also plausible to expect that both types of cross-modal attentional spreading would be facilitated, rather than either inhibited or intact, during

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attentional blink. The present study found that both the stimulus-driven and representation-driven cross-modal spread of attention were completely suppressed during the attentional blink interval but were highly prominent outside the attentional blink interval. These findings demonstrate for the first time that in addition to the previously proposed space-based and object-based visual attention operating at the perceptual stage of processing, adequate post-perceptual ongoing attentional resources are essential another vital endogenous prerequisites for the occurrences of not only the stimulus-driven, but also the representation-driven both types of attentional spreading from visual to auditory modality.

2 | METHOD

2.1 | Participants

A total of 38 healthy subjects (24 females and 14 males; age range of 18–24 years, mean age 20.45 years; all right-handed) participated in this experiment after giving written informed consent as approved by the Human Research Protections Program of SooChow University. Our target sample size was 30. We ran an additional eight participants who were recruited before we reached our target sample size, resulting in a final sample of 38 participants. Experimental procedures were in agreement with the Declaration of Helsinki. All subjects reported normal or corrected-to-normal vision as well as normal hearing, and were naive as to the hypothesis of the experiment.

2.2 | Apparatus, stimuli, and experimental design

The experiment was performed in a dark and sound-attenuated chamber. Stimulus presentation was scripted using "Presentation" software (version 18.0, Neurobehavioral Systems, Inc.). Visual stimuli were presented on a 27-inch LCD monitor (ASUS PG279Q, resolution 1,920 × 1,080, refresh rate 120 Hz) on which the background color was set to gray, and auditory stimuli (if presented) were delivered by a pair of loudspeakers (HiVi X3) positioned at the left and right sides of the monitor. Subjects sat in front of the monitor with a viewing distance of approximately 80 cm, and were required to maintain their eyes fixated on a red cross (.3° × .3° of visual angle), which was displayed at the center of the screen throughout each RSVP stream. The visual stimuli in the experiment consisted of 48 black-and-white line drawings (each 5.6° × 4.5° in visual angle), including 30 unique drawings of houses used as distractors, nine unique drawings (three clothes, three cups, and three flowers) used as the first target (T1), and the remaining nine unique drawings (three dogs, three cars, and three drums) used as the second target (T2). The line drawings for T1 and T2 were from two nonoverlapping sets in order to avoid priming (Koelewijn et al., 2008) or repetition blindness effects (Kanwisher, 1987). The auditory stimuli were comprised of nine unique natural sounds (three barks of dogs, three beeps of cars, and three beats of drums; all stereo) that were 200 ms in duration (with 20 ms rise and fall ramps) and approximately 75 dB in loudness at subjects' ears when delivered. These line drawings and natural sounds were taken from the study of Zhao et al. (2020) wherein similar (but not identical) behavioral task was performed, and all line drawings used in the present study were shown in Figure S1.

Each trial began with the presentation of fixation for a fixed period of 1,000 ms, immediately followed by an RSVP stream presented at the center of the screen (Figure 1a). The RSVP stream was comprised of 17 distinct line drawings, including T1, T2, and 15 distractors [distractors were selected randomly (without repetition) from the 30 drawings of houses for each trial], and each line drawing was presented for 100 ms with a stimulus-onset asynchrony (SOA) of 100 ms. The T1 could be one of the nine drawings (three clothes, three cups, and three flowers) with equal probability, and was presented randomly from the third to the fifth position in the RSVP stream. The T2 was presented either three positions after T1 (i.e., at **lag 3**, T1-to-T2 SOA of 300 ms; 50% of the trials) or eight positions after T1 (i.e., at **lag 8**, T1-to-T2 SOA of 800 ms; 50% of the trials), and could be one of another nine drawings with equal probability (three dogs, three cars, and three drums; 69.231% of the trials) or be blank (i.e., a white rectangle with the same size of line drawings; 30.769% of the trials). The reasons for including the "blank T2" trials were detailed in the Data analysis section (see below).

Along with the RSVP stream, a task-irrelevant natural sound [i.e., one of the nine unique sounds (three barks of dogs, three beeps of cars, and three beats of drums) with equal probability] was delivered synchronously with T2 on 46.154% of the trials, which could be either semantically congruent with T2 (audiovisual congruent condition, labeled as **VAcon**, 23.077% of the trials; e.g., bark with dog) or incongruent with T2 (audiovisual incongruent condition, labeled as **VAincon**, 23.077% of the trials; e.g., beep with dog). Of note, incongruent drawings and sounds were consistently paired throughout the experiment: drawings of dogs were presented with beeps of cars, drawings of cars were presented with beats of drums, and drawings of drums were presented with barks of dogs, in order to test whether the temporary associations formed by repeated audiovisual pairings during the experiment could account for the representation-driven attentional spreading (c.f. Fiebelkorn, Foxe, & Molholm, 2010). In addition, the sound could be delivered alone (i.e., synchronously with the "blank T2") on 23.077% of the trials (auditory-only condition, labeled as **A-only**). On another 23.077% of the trials, T2

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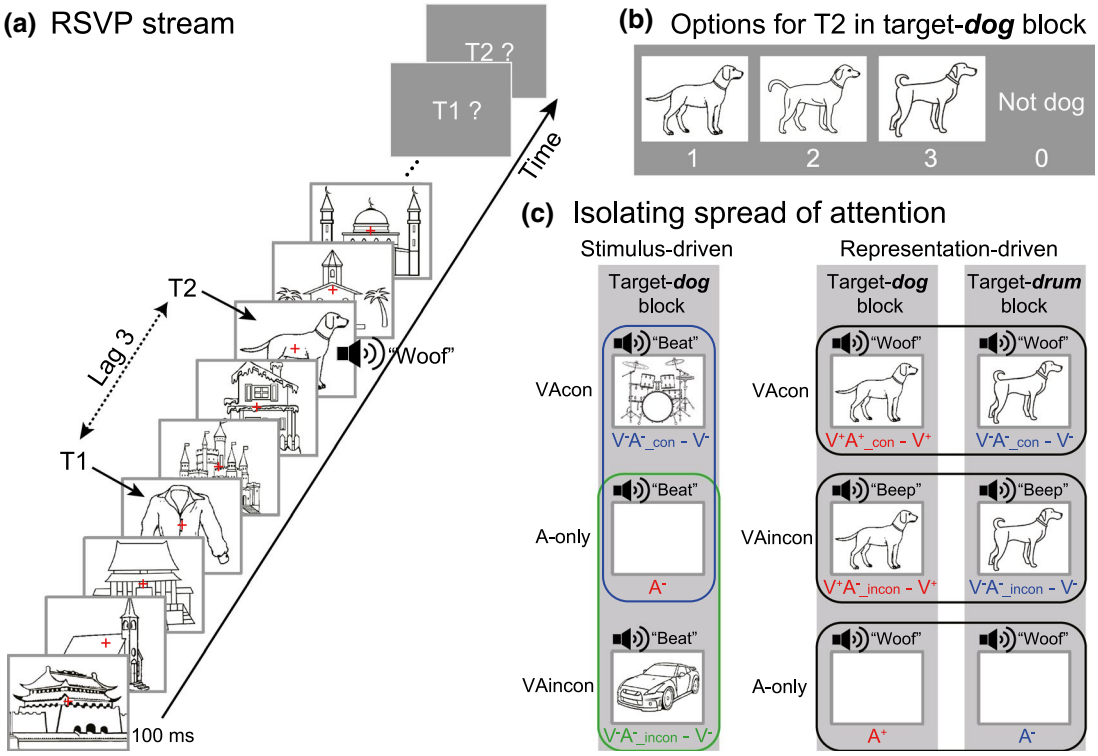

FIGURE 1 (a) Schematic illustration of the RSVP stream for a trial on which T2 was presented three positions after T1 (i.e., at lag 3) and on which a semantically congruent sound was played simultaneously with T2. Each line drawing in the RSVP stream was presented for 100 ms. On a given trial, T2 could be presented either at lag 3 or at lag 8 and could be one of the nine drawings or blank. A task-irrelevant natural sound could be delivered synchronously with the presented T2, alone (with the blank T2), or be absent (with the presented T2 or blank T2). When synchronous with the presented T2, the sound could be either semantically congruent or incongruent with T2 (e.g., bark with dog, or bark with drum). The task for participants was to discriminate, whereas ignoring all sounds, the exact identity of T1, and the exact identity of T2 *only if* the drawing of T2 belonged to the *target* object category (dog, car, or drum) defined before the beginning of a given block. (b) Four options for T2 exemplified in a target-*dog* block. (c) Examples of the experimental comparisons used to isolate the cross-modal spread of attention that originated from the stimulus-driven and representation-driven processes. The superscripts "+" and "−" denote visual or auditory representations of the *target* and *nontarget* categories, respectively, which applies to all relevant figures and paragraphs in the main text. Note that for analysis purpose, responses to target and nontarget categories were collapsed across the three object categories (dog, car, and drum)

was presented without sound (visual-only condition, labeled as **V-only**). On the remaining 7.692% of the trials, no sound nor visual stimulus was presented (i.e., only the "blank T2" was presented; no-stimulus condition, labeled as **No-stim**). All these types of trials were presented with their given probabilities in a pseudo-randomized order.

For T1 in each RSVP stream, the task for participants was to discriminate the exact identity of T1 as accurately as possible in an unspeeded fashion by pressing one of nine buttons on the keyboard's number pad (1–9) with right hand after each RSVP stream. For T2 in each RSVP stream, in contrast, subjects were instructed before a given block that they should discriminate (also in an unspeeded manner) the exact identity of T2 by pressing one of three buttons on the number pad (1, 2, and 3) ***only when*** the line drawing of T2 belonged to the *predefined* ***target object category*** (e.g., dog), otherwise (i.e., T2 was a car or drum, or was blank) they should press the button "0" (see Figure 1b), whereas ignoring all sounds (if delivered). The button-press for T2 then triggered the next trial. In other words, participants were required to further discriminate the exact identity of T2 only when they subjectively perceived the drawing of T2 as belonging to the target object category. Instead, if participants subjectively perceived the drawing of T2 as *not belonging to* the target object category, or if they did *not perceive* a T2 of target object category at all (because the T2 could be blank), they were simply required to press the button "0." These instructions were utilized to encourage the participants to use a strategy of making a binary "target-or-not" classification rather than merely recognizing the object category of T2, thereby ensuring that the subjects would pay further object-based visual selective attention to T2 of target object category relative to T2 of nontarget object categories. The target object category for T2 was defined before the beginning of a given block, which could be dogs, cars, or drums. If one category of objects (e.g., dogs) were the target object category for T2 during a given block, the remaining categories of objects (i.e., cars and drums) were thus the nontarget T2 object categories during that block.

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The order of target-dog, target-car, and target-drum blocks was pseudo-randomized. Therefore, the present experimental design manipulated three critical within-subject factors consisting of: (1) ***lag*** (lag 3, lag 8); (2) ***T2 stimulus type*** (VAcon, VAincon, V-only, A-only, No-stim); and (3) ***object-based attention to T2*** (T2 of target object category, T2 of nontarget object category). Table 1 summarizes the abbreviation, definition, and example for each specific experimental condition deriving from the current design. The whole experiment comprised of 27 blocks (9 target-dog blocks, 9 target-car blocks, 9 target-drum blocks) of 78 trials each. All participants were required to complete three blocks of practice at the beginning of the experiment (a target-dog block, a target-car block, and a target-drum block) in order to get familiar with the stimulus materials and task demands. Participants were allowed to have a rest between blocks in order to relieve fatigue.

2.3 | Electrophysiological recording and preprocessing

The electroencephalogram (EEG) was acquired continuously using a NeuroScan Synamps system (NeuroScan, Inc.) and was recorded from 57 tin electrodes mounted in an elastic cap (Electro-Cap International, Inc.). These electrode sites were positioned according to a modified 10–10 system montage (for details, see McDonald et al., 2003). All EEG electrodes were referenced to an electrode on the left mastoid during data acquisition. Horizontal eye movements were monitored via a bipolar pair of electrodes positioned at the left and right outer canthi (horizontal electrooculogram, HEOG). Vertical eye movements and blinks were monitored bipolarly by two electrodes above and below the left eye (VEOG). The impedances of all electrodes were kept below 5 kΩ. The online EEG and EOG signals were amplified with a gain of 10,000, a band-pass filter of .05–100 Hz, and were continuously digitized with a sampling rate of 1,000 Hz.

In offline processing, the continuous EEG signals were first down-sampled to 500 Hz, and then, low-pass filtered digitally (half-amplitude cutoff = 30 Hz, slope = 24 dB/octave) using a noncausal, zero-phase shifted (two-pass forward and reverse) and squared Butterworth finite impulse response (FIR) filter to attenuate high-frequency noise triggered by muscle activities or external electrical sources. The filtered EEG data were re-referenced to the algebraic mean of the left and right mastoid electrodes. EEG signals were then divided into 600-ms

TABLE 1 Nomenclature, definition, and example for each experimental condition in lag 3 condition, which also applies to those in lag 8 condition

Stimulus type	Target condition	Label	Definition and example
VAcon	Target	$V^{+}A^{+}{}_{_con}$	a T2 belonging to the target category is paired with a semantically congruent sound (e.g., a dog paired with a bark under a target-dog block)
	Nontarget	$V^{-}A^{-}{}_{_con}$	a T2 belonging to the nontarget categories is paired with a semantically congruent sound (e.g., a dog paired with a bark under a target-car or target-drum block)
VAincon	Target	$V^{+}A^{-}{}_{_incon}$	a T2 belonging to the target category is paired with a semantically incongruent sound (e.g., a dog paired with a beep under a target-dog block)
	Nontarget	$V^{-}A^{-}{}_{_incon}$	a T2 belonging to one nontarget category is paired with a semantically incongruent sound belonging to another nontarget category (e.g., a dog with a beep under a target-drum block)
	Nontarget	$V^{-}A^{+}{}_{_incon}$	a T2 belonging to one nontarget category is paired with a semantically incongruent sound belonging to the target category (e.g., a dog paired with a beep under a target-car block)
V-only	Target	V^{+}	a T2 belonging to the target category is paired with no sound (e.g., a dog paired with no sound under a target-dog block)
	Nontarget	V^{-}	a T2 belonging to the nontarget categories is paired with no sound (e.g., a dog paired with no sound under a target-car or target drum block)
A-only	Target	A^{+}	a "blank T2" is paired with a sound belonging to the target category (e.g., a blank drawing paired with a bark under a target-dog block)
	Nontarget	A^{-}	a "blank T2" is paired with a sound belonging to the nontarget categories (e.g., a blank drawing paired with a bark under a target-car or target-drum block)
No-stim	----------	No-stim	a "blank T2" is paired with no sound

Note: (1) the superscripts "+" and "−" denote visual or auditory representations of the *target* and *nontarget* categories, respectively; (2) "----------": not available.

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epochs time-locked to the onset of T2 with a 100-ms prestimulus (T2) baseline and were baseline-corrected. Automatic artifact rejection was performed according to a threshold of ±75 μV for both EEG and EOG channels, in order to eliminate epochs contaminated by eye movements, eye blinks, and muscle activities. On average, these artifacts led to the rejection of 7.8 ± .7% (*M* ± *SE*) of epochs. In addition, according to previous EEG studies using attentional blink paradigms (e.g., Haroush et al., 2011; Kranczioch et al., 2007; Kranczioch & Thorne, 2015; Maier & Rahman, 2018; Vogel et al., 1998), only trials (epochs) on which T1 identity was correctly discriminated were further analyzed. The remaining EEG epochs, sorted according to lag (lag 3, lag 8), stimulus type (VAcon, VAincon, V-only, A-only, No-stim), and target condition (target, nontarget), were averaged to obtain corresponding ERP waveforms. Note that for nontarget VAincon conditions, only trials (epochs) on which their auditory elements' representations were also corresponding to nontarget object category [i.e., $V^{-}A^{-}{}_{_incon}$ condition; e.g., dogs with beeps under target-drum blocks, see Table 1 for reference] were further averaged for ERP analysis (c.f. Fiebelkorn, Foxe, & Molholm, 2010). EEG preprocessing and ERP analysis were carried out using "Scan" software (version 4.5, NeuroScan, Inc.).

After excluding trials with artifacts and incorrect T1 responses, (1) there were on average 62.5 (range 46–76) and 132.0 (range 93–152) valid epochs left on target VAcon and nontarget VAcon trials in lag 3 condition, and 63.1 (range 45–77) and 131.3 (range 92–151) valid epochs left on target VAcon and nontarget VAcon trials in lag 8 condition; (2) there were on average 62.9 (range 37–78) and 66.6 (range 42–82) epochs on target and nontarget VAincon trials in lag 3 condition, and 63.2 (range 38–79) and 65.3 (range 41–81) epochs on target and nontarget VAincon trials in lag 8 condition; (3) there were on average 63.6 (range 41–77) and 132.0 (range 92–153) epochs on target and nontarget V-only trials in lag 3 condition, and 62.4 (range 40–76) and 132.4 (range 91–152) epochs on target and nontarget V-only trials in lag 8 condition; (4) there were on average 66.3 (range 51–77) and 130.9 (range 84–154) epochs on A-only trials with auditory representations of target categories and nontarget categories in lag 3 condition, respectively, and 65.7 (range 50–76) and 130.3 (range 83–154) epochs on A-only trials with auditory representations of target categories and nontarget categories in lag 8 condition, respectively; (5) there were on average 66.2 (range 44–77) valid epochs on No-stim trials in lag 3 condition, and 66.8 (range 45–78) valid epochs on No-stim trials in lag 8 condition.

2.4 | Data analysis

Prior to data analysis, the time-locked ERPs elicited by the No-stim trials (i.e., blank T2 with no sound) were first subtracted from each of the remaining original (VAcon, VAincon, V-only and A-only) ERP waveforms separately for lag 3 and lag 8 conditions, in order to cancel out the systematic superposition of ERPs elicited by the pre- and post-T2 distractors (c.f. Kranczioch & Thorne, 2015; Luo et al., 2010, 2013; Maier & Rahman, 2018; Sergent et al., 2005; Vogel et al., 1998). However, we noted that the blank T2s in No-stim and A-only conditions might additionally elicit some unexpected visual ERPs, and this possibility could not be ruled out under the current experiment design. Nevertheless, these unexpected ERPs in No-stim and A-only conditions, if there were, would be ultimately counterbalanced in any way when further isolating the ERP correlates of either the stimulus-driven or the representation-driven attentional spreading.

To isolate the ***stimulus-driven*** *spread of attention*, the extracted auditory ERPs to audiovisual stimuli when their visual components were drawings of the visual *nontarget* object categories [i.e., $(V^{-}A^{-}{}_{_con} - V^{-})$ or $(V^{-}A^{-}{}_{_incon} - V^{-})$] were compared within blocks with ERPs to auditory-only stimuli when their representations were corresponding to the visual *nontarget* object categories (i.e., A^{-}) (c.f. Fiebelkorn, Foxe, & Molholm, 2010; Fiebelkorn, Foxe, Schwartz, et al., 2010). Specifically, for "***congruent***" audiovisual stimuli, for example (Figure 1c, left, 1st & 2nd rows), the extracted auditory ERPs to beats of drums presented with drawings of drums under target-dog blocks $(V^{-}A^{-}{}_{_con} - V^{-})$ were compared within blocks with auditory-only ERPs to beats of drums (A^{-}). For "***incongruent***" audiovisual stimuli, for instance (Figure 1c, left, 2nd & 3rd rows), the extracted auditory ERPs to beats of drums presented with drawings of cars under target-dog blocks $(V^{-}A^{-}{}_{_incon} - V^{-})$ were compared within blocks with auditory-only ERPs to beats of drums (A^{-}). These comparisons were conducted separately for lag 3 and lag 8 conditions. Note that subtracting the visual-only ERPs from the audiovisual ERPs allows for the extraction of auditory ERPs along with any possible cross-modal spread of attention effect (e.g., Busse et al., 2005; Donohue et al., 2011; Fiebelkorn, Foxe, & Molholm, 2010; Fiebelkorn, Foxe, Schwartz, et al., 2010; Molholm et al., 2007; Zimmer et al., 2010). In addition, because the present behavioral task required subjects to distinguish the target T2s from nontarget T2s, and then, discriminate the exact identities of target T2s whereas ignoring all sounds, the visual modality was always attended (although the target T2s would receive further object-based attention relative to nontarget T2s) and the auditory modality was always unattended.

The ***representation-driven*** *spread of attention* was isolated as follows. (1) For ***audiovisual*** stimuli, the extracted auditory ERPs to audiovisual stimuli when their visual components were drawings of the visual *target* object categories [i.e., $(V^{+}A^{+}{}_{_con} - V^{+})$ or $(V^{+}A^{-}{}_{_incon} - V^{+})$] were compared across blocks with the extracted auditory ERPs to the same audiovisual stimuli when their visual components were drawings of

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the visual *nontarget* object categories [i.e., ($V^-A^-_{_con} - V^-$) or ($V^-A^-_{_incon} - V^-$)] (c.f. Fiebelkorn, Foxe, & Molholm, 2010; Molholm et al., 2007). Specifically, for "***congruent***" audiovisual stimuli, for example (Figure 1c, right, 1st row), the extracted auditory ERPs to barks of dogs presented with drawings of dogs under target-dog blocks ($V^+A^+_{_con} - V^+$) were compared across blocks with the extracted auditory ERPs to the same stimuli under either target-car or target-drum blocks ($V^-A^-_{_con} - V^-$). For "***incongruent***" audiovisual stimuli, for instance (Figure 1c, right, 2nd row), the extracted auditory ERPs to beeps of cars presented with drawings of dogs under target-dog blocks ($V^+A^-_{_incon} - V^+$) were compared across blocks with the extracted auditory ERPs to the same stimuli under target-drum blocks ($V^-A^-_{_incon} - V^-$). (2) For ***auditory-only*** stimuli (Figure 1c, right, 3rd row), ERPs to auditory stimuli when their representations were corresponding to the visual *target* object categories (A^+; e.g., barks of dogs under target-dog blocks) were compared across blocks with ERPs to the same auditory stimuli when their representations were corresponding to the visual *nontarget* object categories (A^-; e.g., barks of dogs under either target-car or target-drum blocks) (c.f. Fiebelkorn, Foxe, & Molholm, 2010; Molholm et al., 2007). These comparisons were also conducted separately for lag 3 and lag 8 conditions.

The present study measured both the stimulus-driven and representation-driven spread of attention effects by the mean amplitude of ***Nd component***, using a time window of 200–300 ms after sound onset and an anterior region of interest (ROI) consisting of nine neighboring fronto-central electrodes (F1, Fz, F2, FC1, FCz, FC2, C1, Cz, C2). These time window and electrodes were chosen a priori based on the study of Fiebelkorn, Foxe, and Molholm (2010) wherein both types of attentional spreading were quantified as mean amplitudes within 200–300 ms interval over medial fronto-central electrodes. Moreover, because many previous studies have found that the stimulus-driven spread of attention was also highly prominent, or even greater, after 300 ms (Busse et al., 2005; Donohue et al., 2011; Fiebelkorn, Foxe, & Molholm, 2010; Fiebelkorn, Foxe, Schwartz, et al., 2010; Talsma et al., 2007; Zimmer et al., 2010), the mean amplitude of Nd component was also measured within a time window of 300–400 ms over the fronto-central ROI in the current study. In addition, since the behavioral task manipulated the object-based visual attention to T2, the selection negativity (SN) was expected to occur when comparing the target with nontarget V-only trials. Thus, the ***SN component*** was quantified as the mean voltage within a time interval of 240–290 ms over a posterior ROI comprised of eight bilaterally parieto-occipital sites (P7, P5, PO7, PO3; P6, P8, PO4, PO8), in line with the extant literature about the timing and topographical distribution of SN (Harter & Aine, 1984; Hillyard & Anllo-Vento, 1998; Molholm et al., 2004, 2007; Smid et al., 1999; Talsma et al., 2007).

To examine whether subjects maintained their object-based visual attention to T2 as instructed, a two-way repeated-measure ANOVA with factors of target condition (target, nontarget) and lag (lag 3, lag 8) was conducted on the mean amplitude of SN component for V-only conditions. Subsequently, to test whether the ***stimulus-driven spread of attention*** would be modulated by audiovisual congruency and the amount of post-perceptual attentional resources that were manipulated via the attentional blink task, a two-way repeated-measure ANOVA with factors of stimulus type [(VAcon − V), (VAincon − V), A-only] and lag (lag 3, lag 8) was conducted on the mean amplitudes within the Nd intervals (200–300 ms and 300–400 ms after sound onset) for ERP waveforms elicited by nontarget object categories (note for A-only condition: ERP waveforms elicited by auditory representations of nontarget object categories). *P*-values for ANOVA results were corrected by the Greenhouse–Geisser method when the sphericity assumption was violated. Pairwise comparisons were made only after finding a significant main effect of stimulus type. Furthermore, to replicate the ***representation-driven spread of attention*** effect and to examine whether it would be modulated by post-perceptual attentional resources, a three-way repeated-measure ANOVA with factors of audiovisual congruency [(VAcon − V), (VAincon − V)], target condition (target, nontarget) and lag (lag 3, lag 8) was conducted on the mean amplitudes within the Nd intervals (200–300 ms and 300–400 ms). Lastly, a separate two-way repeated-measure ANOVA with factors of target condition (target, nontarget) and lag (lag 3, lag 8) was performed on the mean amplitude within the Nd intervals (200–300 ms and 300–400 ms) for A-only conditions, in order to examine whether the representation-driven spread of attention effect is independent of the co-occurrence of visual and auditory stimuli and whether it would be modulated by post-perceptual attentional resources. To control the inflation of Type I error rate, *p*-values for all follow-up analyses (if necessary) of each omnibus ANOVA above were adjusted using the false discovery rate (FDR) correction (Benjamini & Hochberg, 1995), and the FDR-corrected *p*-value was denoted as "p_{FDR}."

3 | RESULTS

3.1 | Behavioral results

The T2 ***discrimination*** accuracy (given correct T1 discrimination) for T2_target trials (Figure 2a, left graph; a response was coded as "correct" *only if* the exact identity of a target T2 was correctly discriminated) was subject to a two-way repeated-measure ANOVA with factors of stimulus type (V, VAcon, VAincon) and lag (lag 3, lag 8). The main effect of lag was significant [$F(1, 37) = 191.13$,

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$p < .0001$, $\eta^2_p = .84$], with much lower T2 discrimination accuracy at lag 3 [45.5 ± 2.8% ($M \pm SE$)] than lag 8 (66.8 ± 2.4%), indicative of the classic attentional blink effect (Raymond et al., 1992). Besides, the main effect of stimulus type [$F(2, 74) = 3.44$, $p = .037$, $\eta^2_p = .09$] and the stimulus type × lag interaction [$F(2, 74) = 12.81$, $p < .0001$, $\eta^2_p = .26$] were also significant. Specific comparisons, however, showed that significantly lower T2 discrimination accuracy at lag 3 than lag 8 (i.e., the attentional blink) was robust not only in the unimodal V condition [lag 3:42.9 ± 2.8%; lag 8:68.6 ± 2.5%; $F(1, 37) = 209.59$, $p_{FDR} < .0001$, $\eta^2_p = .85$], but also in the bimodal VAcon condition [lag 3:47.5 ± 2.8%; lag 8:67.3 ± 2.3%; $F(1, 37) = 114.90$, $p_{FDR} < .0001$, $\eta^2_p = .76$] and VAincon condition [lag 3:46.2 ± 3.0%; lag 8:64.6 ± 2.8%; $F(1, 37) = 114.84$, $p_{FDR} < .0001$, $\eta^2_p = .76$]. These results are consistent with previous findings that the attentional blink was robust even when a sound was presented synchronously with the visual T2 (Adam & Noppeney, 2014; Haroush et al., 2011; Kranczioch & Thorne, 2013, 2015; Olivers & Van der Burg, 2008; Zhao et al., 2020), thereby demonstrating that the attentional blink paradigm is suitable for manipulating attentional resources under the experimental design of cross-modal attentional spreading.

The same two-way ANOVA was further performed on the T2 ***classification*** accuracy (given correct T1 discrimination) for T2_target trials (Figure 2a, right graph; note here a correct response was re-defined as any button-press among 1, 2, and 3 in response to a target T2 *regardless of* its exact identity, see Figure 1b). There was also a significant main effect of lag [$F(1, 37) = 91.83$, $p < .0001$, $\eta^2_p = .71$], with much lower T2 classification accuracy at lag 3 (75.6 ± 2.6%) than lag 8 (90.8 ± 1.5%). Besides, the main effect of stimulus type [$F(2, 74) = 12.05$, $p < .0001$, $\eta^2_p = .25$] and the stimulus type × lag interaction [$F(2, 74) = 8.58$, $p = .0004$, $\eta^2_p = .19$] were also significant. Similarly, planned comparisons yielded that the lower T2 classification accuracy at lag 3 than lag 8 was found not only in V condition [lag 3:72.7 ± 2.9%; lag 8:90.9 ± 1.6%; $F(1, 37) = 74.62$, $p_{FDR} < .0001$, $\eta^2_p = .67$], but also in VAcon condition [lag 3:78.9 ± 2.5%; lag 8:91.8 ± 1.4%; $F(1, 37) = 72.74$, $p_{FDR} < .0001$, $\eta^2_p = .66$] and VAincon condition [lag 3:75.0 ± 2.6%; lag 8:89.6 ± 1.7%;

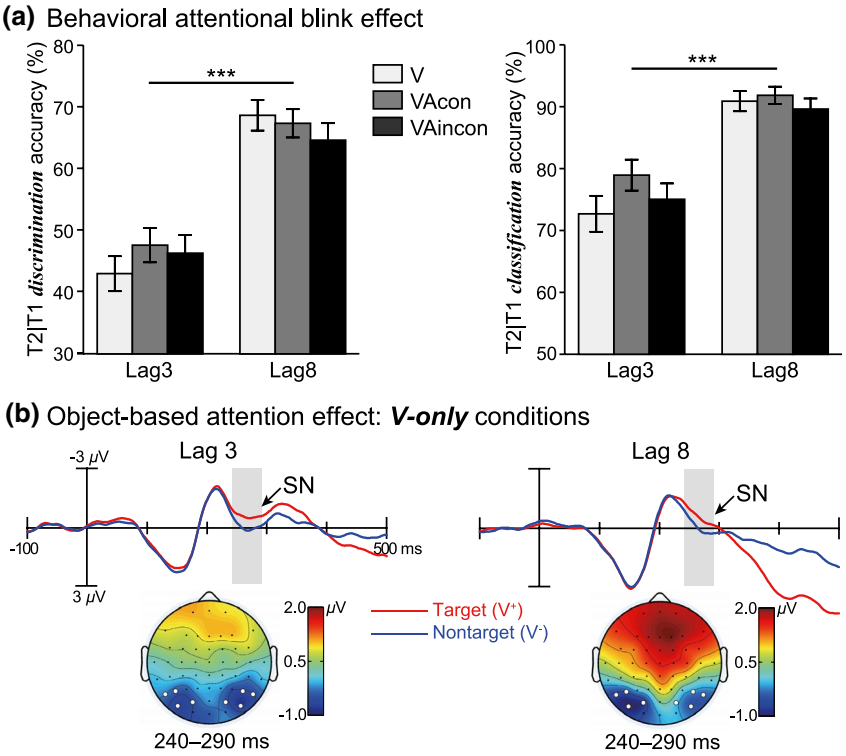

FIGURE 2 (a) T2 ***discrimination*** accuracy (left graph) and T2 ***classification*** accuracy (right graph) for T2 of target object category as functions of lag (lag 3, lag 8) and stimulus type (V-only, VAcon, VAincon). A reliable attentional blink effect was found, manifested as either significantly lower T2 discrimination accuracy or significantly lower T2 classification accuracy at lag 3 than lag 8. Error bars represent ± 1 *SE*; ***$p < .001$. (b) ERP waveforms elicited by V-only trials as functions of lag (lag 3, lag 8) and target condition (target, nontarget), which were averaged over the posterior ROI (P7, P5, PO7, PO3; P6, P8, PO4, and PO8). The shaded areas on waveforms depict the time window (240–290 ms) within which the SN component was quantified. Scalp topographies are shown for the target minus nontarget difference amplitude within the SN interval. The white dots on scalp topographies depict the posterior ROI over which the SN component was quantified. The SN component, indexed by significantly larger negative amplitude on target than nontarget trials, was evident regardless of lag, demonstrating a robust object-based visual attention effect

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$F(1, 37) = 83.05$, $p_{FDR} < .0001$, $\eta^2_p = .69$]. Thus, the reliable attentional blink effect observed also on T2 classification accuracy provides strong evidence that, at least in the current study, the post-perceptual processing of information regarding both the *detailed T2 identity* and the *abstract T2 object category* was impaired during the attentional blink interval.

In addition, to provide evidence that participants had successfully ignored the task-irrelevant sounds as instructed, the false alarms of T2 (i.e., button-press of 1, 2, or 3; see Figure 1b) to auditory-alone representations of the visual target T2 (i.e., A^+ condition) and auditory representations of the visual target paired with incongruent drawings of nontarget T2 (i.e., $V^-A^+_{_incon}$ condition) were quantified. These errors, if frequent, would suggest that participants were unable to ignore the auditory modality (c.f. Fiebelkorn, Foxe, & Molholm, 2010). The results showed that the T2 false alarm rates (given correct T1 discrimination) in A^+ condition were only 4.8 ± 1.0% and 4.6 ± 1.0% at lag 3 and lag 8, respectively, and the T2 false alarm rates in $V^-A^+_{_incon}$ condition were only 2.2 ± .6% and 1.7 ± .6% at lag 3 and lag 8, respectively. These negligible incidences of false alarms demonstrate that the auditory modality was indeed unattended in the present study.

3.2 | ERP results

3.2.1 | Evidence of object-based visual selective attention

To check for subjects' object-based visual selective attention to T2, a two-way repeated-measure ANOVA with factors of target condition (target, nontarget) and lag (lag 3, lag 8) was conducted on the mean amplitudes within the SN interval (240–290 ms after T2 onset) over the bilateral parieto-occipital ROI for ERP waveforms in V-only conditions (Figure 2b). The results showed a significant main effect of target condition [$F(1, 37) = 19.37$, $p < .0001$, $\eta^2_p = .34$], with larger negative amplitude in response to target than nontarget object categories [target minus nontarget: −.52 ± .12 μV ($M \pm SE$)], indicating the presence of the SN difference component (Harter & Aine, 1984; Hillyard & Anllo-Vento, 1998; Molholm et al., 2004, 2007; Smid et al., 1999). However, neither the main effect of lag [$F(1, 37) = .03$, $p = .875$, $\eta^2_p = .001$] nor the target condition × lag interaction [$F(1, 37) = .02$, $p = .904$, $\eta^2_p = .0004$] was significant. These findings not only demonstrate that participants maintained their object-based visual attention to T2 as instructed, but also confirm our prediction that the attentional blink with a post-perceptual locus of suppression would not impair the perceptual processing necessary for object-based attentional selection.

A supplementary analysis of the P3b component, which is a well-accepted ERP signature of the behavioral attentional blink effect (Sergent et al., 2005; Vogel et al., 1998), was shown in the caption to Figure S2.

3.2.2 | The stimulus-driven spread of attention is dependent on post-perceptual attentional resources but independent of audiovisual congruency

To test whether the stimulus-driven spread of attention would be modulated by audiovisual congruency and post-perceptual attentional resources, a two-way repeated-measure ANOVA with factors of stimulus type [($V^-A^-_{_con} - V^-$), ($V^-A^-_{_incon} - V^-$), A^-] and lag (lag 3, lag 8) was conducted on the mean amplitudes within the Nd intervals (200–300 ms and 300–400 ms after sound onset) over the fronto-central ROI, which was limited to ERP waveforms elicited by *nontarget* object categories (Figure 3). There was a significant stimulus type × lag interaction within both 200–300 ms interval [$F(2, 74) = 10.75$, $p < .0001$, $\eta^2_p = .23$] and 300–400 ms interval [$F(2, 74) = 14.81$, $p < .0001$, $\eta^2_p = .29$], suggesting the main effect of stimulus type might be affected by lag. To investigate this interaction, one-way ANOVAs with a single factor of stimulus type were performed separately for lag 8 and lag 3 conditions.

As follows from above, the main effect of stimulus type was significant in lag 8 condition within both time windows [200–300 ms: $F(2, 74) = 17.86$, $p_{FDR} < .0001$, $\eta^2_p = .33$; 300–400 ms: $F(2, 74) = 24.18$, $p_{FDR} < .0001$, $\eta^2_p = .40$]. Pairwise comparisons in lag 8 condition showed that the stimulus-driven Nd components, indexed by significantly greater negative amplitudes on the extracted auditory than auditory-only ERP waveforms, were evident for both congruent and incongruent audiovisual nontarget stimuli during 200–300 ms interval [both $p_{FDR} < .0001$, $d > .83$; ($V^-A^-_{_con} - V^-$) − A^- = −.93 ± .19 μV; ($V^-A^-_{_incon} - V^-$) − A^- = −1.13 ± .15 μV] and 300–400 ms interval [both $p_{FDR} < .0002$, $d > .79$; ($V^-A^-_{_con} - V^-$) − A^- = −1.08 ± .24 μV; ($V^-A^-_{_incon} - V^-$) − A^- = −1.34 ± .24 μV; Figure 3b; Figure 5a]. Moreover, the Nd amplitudes did not differ significantly between incongruent and congruent audiovisual nontarget stimuli during both intervals [200–300 ms: $p_{FDR} = .338$, $d = .18$; 300–400 ms: $p_{FDR} = .273$, $d = .18$]. In contrast, the main effect of stimulus type in lag 3 condition was in fact nonsignificant within either time interval [200–300 ms: $F(2, 74) = .23$, $p_{FDR} = .757$, $\eta^2_p = .01$; 300–400 ms: $F(2, 74) = 1.85$, $p_{FDR} = .205$, $\eta^2_p = .05$], indicating the Nd components for both congruent and incongruent audiovisual nontarget object categories were absent under the attentional blink [200–300 ms: ($V^-A^-_{_con} - V^-$) − A^- = .08 ± .27 μV, ($V^-A^-_{_incon} - V^-$) − A^- = .05 ± .29 μV; 300–400 ms: ($V^-A^-_{_con} - V^-$) − A^- = .32 ± .24 μV, ($V^-A^-_{_incon} - V^-$) − A^- = .43 ± .23 μV; Figure 3a; Figure 5a]. Therefore, these findings

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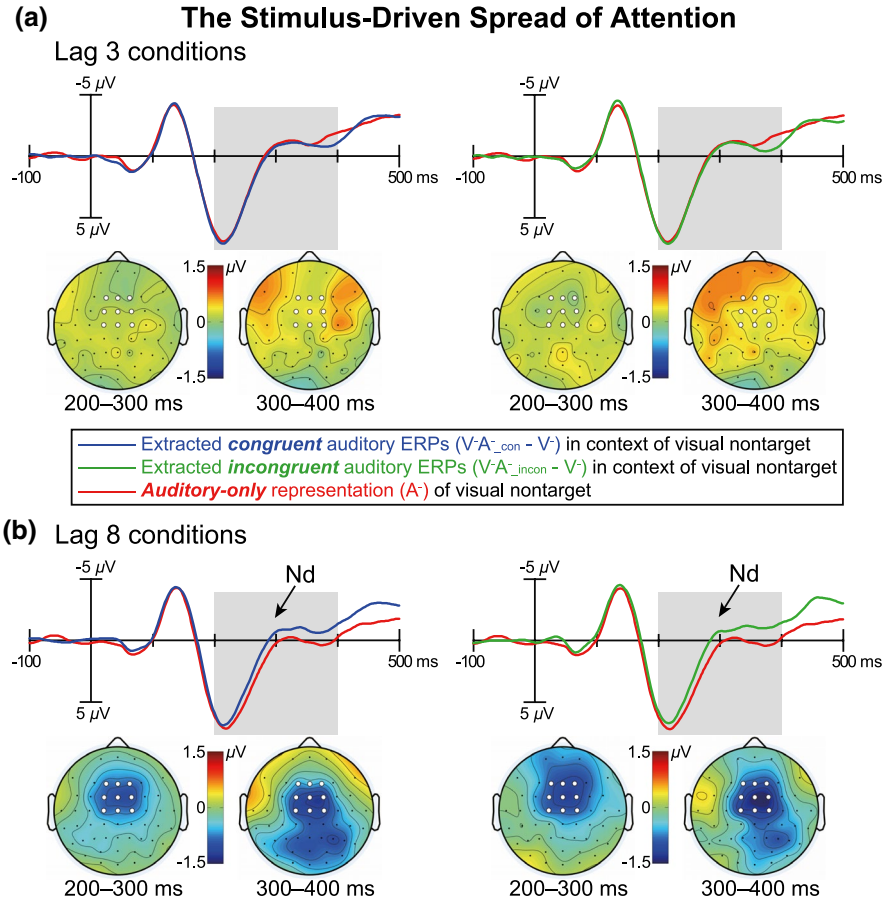

FIGURE 3 The extracted auditory ERP waveforms to *congruent* (blue traces) and *incongruent* (green traces) audiovisual stimuli when their visual components were the nontarget object categories, and ERP waveforms evoked by *A-only* stimuli (red traces) when their representations were corresponding to the nontarget object categories. These ERP waveforms were plotted separately for (a) lag 3 and (b) lag 8 conditions, and were averaged over the fronto-central ROI (F1, Fz, F2, FC1, FCz, FC2, C1, Cz, and C2). The shaded areas on the waveforms depict the time windows (200–300 ms and 300–400 ms) within which the Nd component was quantified. Scalp topographies under each lag condition are shown separately for the $[(V^{-}A^{-}_{con} - V^{-}) - A^{-}]$ and $[(V^{-}A^{-}_{incon} - V^{-}) - A^{-}]$ difference amplitudes during 200–300 ms and 300–400 ms intervals. The white dots on scalp topographies depict the fronto-central ROI over which the Nd component was quantified. The Nd components, indexed by significantly greater negative amplitudes on the extracted auditory than auditory-only ERP waveforms, were evident for both congruent and incongruent audiovisual nontarget stimuli but only at lag 8, indicating the stimulus-driven attentional spreading is dependent on post-perceptual attentional resources but independent of audiovisual congruency

not only replicate the previous observation that the stimulus-driven cross-modal attentional spreading was independent of highly learned audiovisual associations (Fiebelkorn, Foxe, & Molholm, 2010), but also demonstrate that the occurrence of stimulus-driven attentional spreading from visual to auditory modality is also dependent on sufficient post-perceptual attentional resources even though all visual stimuli co-occurring with task-irrelevant sounds are spatially attended.

3.2.3 | The representation-driven spread of attention is also dependent on post-perceptual attentional resources

Given that the visual selection negativity (SN) was not impaired at lag 3, which suggests a sufficient top-down activation of visual targets' representations during the attentional blink interval, it is interesting to examine whether the representation-driven spread of attention would be modulated by attentional blink. A three-way repeated-measure ANOVA with factors of audiovisual congruency [(VAcon − V), (VAincon − V)], target condition (target, nontarget) and lag (lag 3, lag 8) was conducted on the mean amplitudes within the Nd intervals (200–300 ms and 300–400 ms) over the fronto-central ROI (Figure 4a,b). The results showed that there was a significant three-way interaction of congruency × target condition × lag within both 200–300 ms interval [$F(1, 37) = 6.74$, $p = .013$, $\eta^2_p = .15$] and 300–400 ms interval [$F(1, 37) = 24.95$, $p < .0001$, $\eta^2_p = .40$], suggesting the representation-driven Nd component (target minus nontarget difference) might be affected by audiovisual congruency in different ways under lag 3 and lag 8 conditions.

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To investigate this three-way interaction, additional congruency × target condition ANOVAs were performed separately for lag 8 and lag 3 conditions.

The two-way ANOVA under lag 8 condition showed a significant interaction of congruency × target condition within both time intervals [200–300 ms: $F(1, 37) = 8.41$, $p_{FDR} = .015$, $\eta^2_p = .19$; 300–400 ms: $F(1, 37) = 40.82$, $p_{FDR} < .0001$, $\eta^2_p = .53$]. Specific comparisons revealed that the representation-driven Nd component, indexed by significantly larger negative amplitude in response to target than nontarget object categories, was evident only for congruent audiovisual stimuli [200–300 ms: $F(1, 37) = 19.82$, $p_{FDR} = .0004$, $\eta^2_p = .35$, target minus nontarget = −.99 ± .22 μV; 300–400 ms: $F(1, 37) = 36.12$, $p_{FDR} < .0001$, $\eta^2_p = .49$, target minus nontarget = −1.38 ± .23 μV; Figure 4a, lower half] but not for incongruent audiovisual stimuli [200–300 ms: $F(1, 37) = .02$, $p_{FDR} = .882$, $\eta^2_p = .001$, target minus nontarget = −.04 ± .29 μV; 300–400 ms: $F(1, 37) = 2.36$, $p_{FDR} = .178$, $\eta^2_p = .06$, target minus nontarget = .31 ± .21 μV; Figure 4b, lower half; Figure 5b]. In contrast, the same two-way ANOVA but under lag 3 condition reveal neither significant interaction [200–300 ms: $F(1, 37) = .81$, $p_{FDR} = .578$, $\eta^2_p = .02$; 300–400 ms: $F(1, 37) = .01$, $p_{FDR} = .923$, $\eta^2_p = .0003$] nor significant main effect of target condition [200–300 ms: $F(1, 37) = .16$, $p_{FDR} = .861$, $\eta^2_p = .004$; 300–400 ms: $F(1, 37) = 2.25$, $p_{FDR} = .178$, $\eta^2_p = .06$], indicating the representation-driven Nd component was not evoked under the attentional blink [200–300 ms: VAcon, target minus nontarget = .02 ± .28 μV, VAincon, target minus nontarget = −.24 ± .34 μV; 300–400 ms: VAcon, target minus nontarget = .39 ± .29 μV, VAincon, target minus nontarget = .36 ± .29 μV; Figure 4a,b, upper halves; Figure 5b]. These findings are consistent with the previous observation that the representation-driven attentional spreading was dependent on highly learned audiovisual associations (Fiebelkorn, Foxe, & Molholm, 2010), and demonstrate newly that the occurrence of representation-driven spread of attention from visual to auditory modality is also contingent

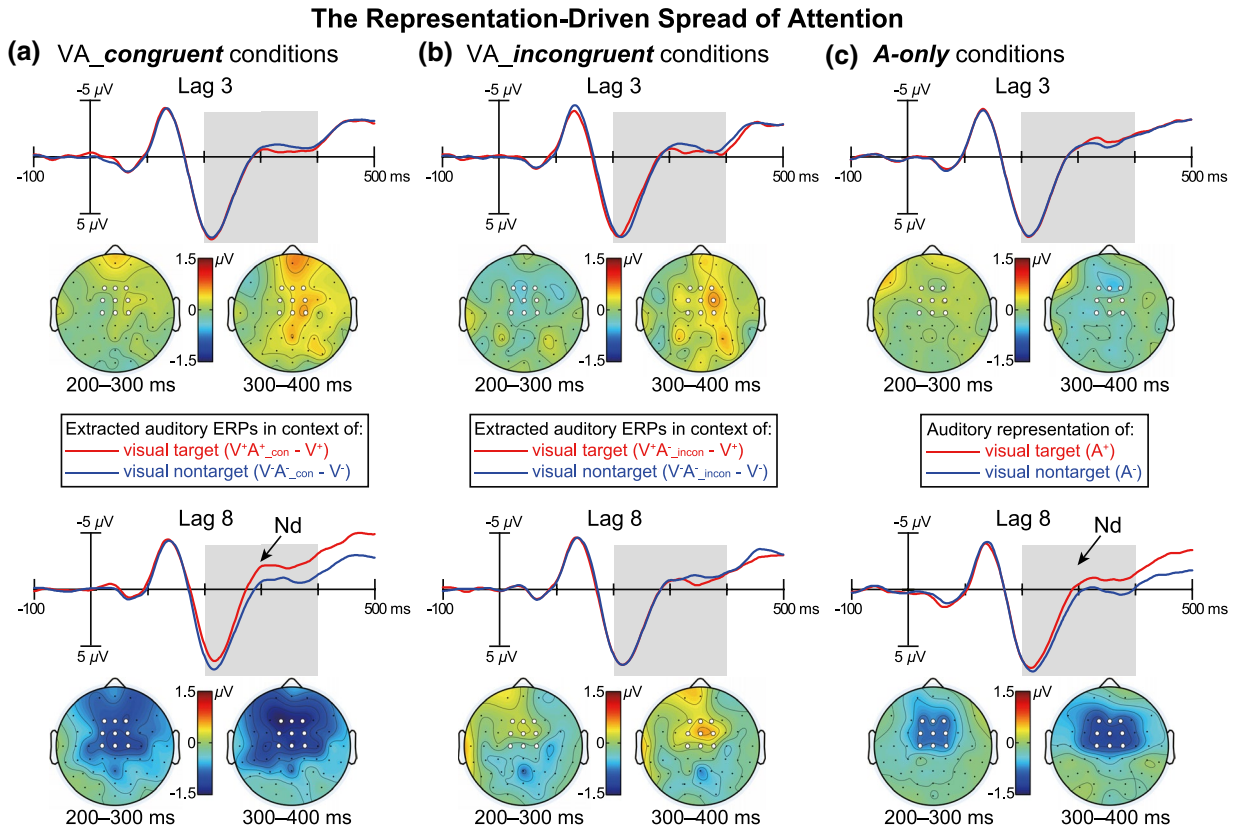

FIGURE 4 The extracted auditory ERP waveforms to audiovisual stimuli when their visual components were the *target* (red traces) and *nontarget* (blue traces) object categories, plotted separately for (a) VAcon and (b) VAincon conditions and for lag 3 (upper halves) and lag 8 (lower halves) conditions. (c) ERP waveforms evoked by A-only stimuli when their representations were corresponding to the visual *target* (red traces) and *nontarget* (blue traces) object categories, plotted separately for lag 3 (upper half) and lag 8 (lower half) conditions. These ERP waveforms were averaged over the same fronto-central ROI as Figure 3. The shaded areas on waveforms depict the same Nd intervals (200–300 ms and 300–400 ms) as Figure 3. Scalp topographies are shown for the target-minus-nontarget difference amplitude within 200–300 ms and 300–400 ms intervals. The Nd components, indexed by significantly larger negative amplitudes on target than nontarget trials, were evident only under VAcon and A-only conditions at lag 8, indicating the representation-driven attentional spreading is dependent on post-perceptual attentional resources and audiovisual congruency but independent of the co-occurrence of visual stimuli

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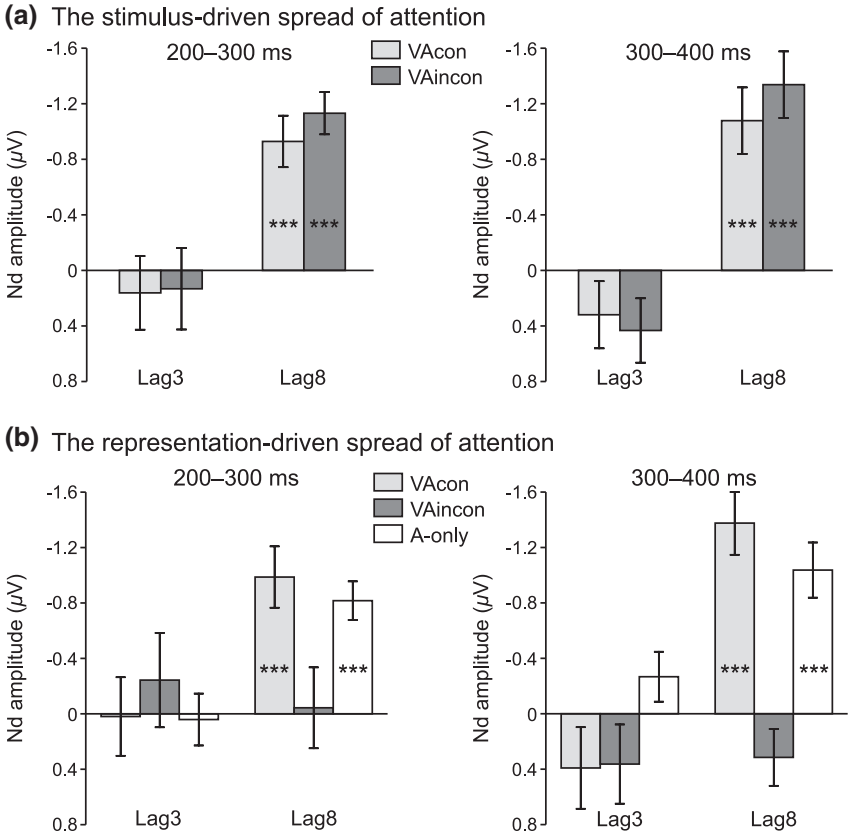

FIGURE 5 A summary for mean amplitudes of (a) the Nd reflecting stimulus-driven spread of attention and (b) the Nd indicating representation-driven spread of attention during the time windows of 200–300 ms (left bars) and 300–400 ms (right bars), averaged over the fronto-central ROI. The stimulus-driven Nd (a) was quantified as the amplitude difference between the extracted auditory and auditory-only nontarget waveforms, whereas the representation-driven Nd (b) was quantified as the target-minus-nontarget difference amplitude. Error bars represent ± 1 *SE*; *** $p_{FDR} < .001$ for a significant stimulus-driven or representation-driven Nd amplitude against zero (after FDR correction). Additional *post hoc* comparisons under lag 8 condition revealed that the representation-driven Nd amplitude did not differ significantly between VAcon and A-only conditions within either 200–300 ms interval [$t(37) = .67$, $p_{FDR} = .508$, $d = .11$] or 300–400 ms interval [$t(37) = 1.09$, $p_{FDR} = .284$, $d = .18$]. In contrast, the representation-driven Nd amplitude was significantly larger in A-only than VAincon conditions during both 200–300 ms interval [$t(37) = 2.34$, $p_{FDR} < .050$, $d = .38$] and 300–400 ms interval [$t(37) = 4.70$, $p_{FDR} < .0001$, $d = .76$]

on sufficient post-perceptual attentional resources even though the top-down activation of visual targets' representations is intact.

Finally, a separate two-way repeated-measure ANOVA with factors of target condition (target, nontarget) and lag (lag 3, lag 8) was performed on the mean amplitudes during the Nd intervals (200–300 ms and 300–400 ms) over the fronto-central ROI for ERP waveforms in A-only conditions (Figure 4c). The results showed a significant target condition × lag interaction within both 200–300 ms interval [$F(1, 37) = 21.17$, $p < .0001$, $\eta^2_p = .36$] and 300–400 ms interval [$F(1, 37) = 10.33$, $p = .003$, $\eta^2_p = .22$]. Specific contrasts revealed that ERP waveforms to auditory-only stimuli were significantly more negative when they were representations of visual targets than when they were representations of visual nontargets only under lag 8 condition [200–300 ms: $F(1, 37) = 34.22$, $p_{FDR} < .0001$, $\eta^2_p = .48$, target minus nontarget = −.82 ± .14 μV; 300–400 ms: $F(1, 37) = 27.12$, $p_{FDR} < .0001$, $\eta^2_p = .42$, target minus nontarget = −1.04 ± .20 μV; Figure 4c, lower half] but not under lag 3 condition [200–300 ms: $F(1, 37) = .05$, $p_{FDR} = .827$, $\eta^2_p = .001$, target minus nontarget = .04 ± .19 μV; 300–400 ms: $F(1, 37) = 2.20$, $p_{FDR} = .146$, $\eta^2_p = .06$, target minus nontarget = −.27 ± .18 μV; Figure 4c, upper half; Figure 5b]. These results are again in agreement with the existing findings that the representation-driven attentional spreading from visual to auditory modality did not necessarily require the co-occurrence of visual stimuli (Fiebelkorn, Foxe, & Molholm, 2010; Molholm et al., 2007), and provide convergent evidence that adequate post-perceptual attentional resources are essential prerequisites for the occurrence of representation-driven spread of attention.

4 | DISCUSSION

The present ERP study, utilizing the attentional blink paradigm to manipulate the amount of post-perceptual attentional

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resources, aimed at further determining whether the long-latency stimulus-driven and representation-driven cross-modal attentional spreading from visual to auditory modality would be restricted or enhanced by the lack of post-perceptual attentional resources, whereas ensuring all visual stimuli are spatially attended and the representations of visual target object categories are activated, which were previously considered to be the only endogenous prerequisites for triggering cross-modal spread of attention (Fiebelkorn, Foxe, & Molholm, 2010; Talsma et al., 2010; Tang et al., 2016). Behaviorally, a robust attentional blink effect (i.e., decreased T2 *discrimination* accuracy for T2s of target object categories at lag 3 than lag 8; Raymond et al., 1992) was found regardless of the presence of the sounds, which is consistent with previous findings that the attentional blink was robust even when a sound was presented synchronously with the visual T2 (Adam & Noppeney, 2014; Haroush et al., 2011; Kranczioch & Thorne, 2013, 2015; Olivers & Van der Burg, 2008; Zhao et al., 2020), thereby demonstrating that the attentional blink paradigm is suitable for manipulating attentional resources under the experimental design of cross-modal attentional spreading. Furthermore, the attentional blink effect was found reliable even on the T2 *classification* accuracy, which provides strong evidence that, at least in the current study, the post-perceptual processing of information regarding both the detailed T2 identity and the abstract T2 object category was impaired during the attentional blink interval.

The ERP analysis (time-locked to T2 onset) of visual-only conditions first showed an overall SN difference component (i.e., larger negativity for target than nontarget object categories; Hillyard & Anllo-Vento, 1998; Molholm et al., 2004, 2007; Talsma et al., 2007) over bilaterally parieto-occipital region, providing evidence that subjects maintained their object-based visual selective attention to T2 as instructed. Besides, similar SN amplitude was also found under lag 3 condition, indicating that the T2 identity (or at least its object category) could be recognized to a certain extent during the attentional blink interval, which is in close agreement with many prior studies revealing that the attentional blink does not disrupt the target-detection process (Dux et al., 2014; Nieuwenstein et al., 2005; Olivers & Meeter, 2008; Raffone et al., 2014; Wyble et al., 2009; Zivony & Lamy, 2016). In contrast to the intact object-recognition process, the present study showed that the final conscious reports of both T2 identity and T2 object category were substantially impaired during the attentional blink interval. This disparity is consistent with the existing influential empirical evidence and theories converging on that the attentional blink impairs or delays further consolidation of T2 identity in working memory (Chun & Potter, 1995; Dux & Marois, 2009; Dux et al., 2014; Kranczioch et al., 2007; Olivers et al., 2007; Ptito et al., 2008; Raffone et al., 2014; Sergent et al., 2005; Vogel et al., 1998; Zivony & Lamy, 2016). Therefore, the finding of intact SN component in lag 3 condition provides confirmative evidence for the success of current experimental design: overloading the post-perceptual attentional resources (by attentional blink) in an orthogonal manner with respect to the spatial-based and object-based visual voluntary attention.

After ensuring the successful manipulations of both attentional blink and object-based visual attention to T2, the present study found that under lag 8 condition, both the stimulus-driven and representation-driven spread of attention were prominent (both indexed by the presence of auditory Nd component). During the attentional blink interval (i.e., under lag 3 condition), in sharp contrast, not only the representation-driven, but also the stimulus-driven spread of attention was found to be completely absent. As noted in the Introduction section, the extant literature suggests that the spatial-based and object-based visual selective attention operating at the perceptual stage of processing might be the only ***endogenous*** prerequisites for triggering the stimulus-driven and representation-driven spread of attention, respectively (Busse et al., 2005; Fiebelkorn, Foxe, & Molholm, 2010; Molholm et al., 2007; Talsma et al., 2010; Tang et al., 2016). However, the current study yielded that whereas the visual stimuli co-occurring with task-irrelevant sounds were always spatially attended, the stimulus-driven spread of attention was completely suppressed when the post-perceptual attentional resources were limited by attentional blink. Similarly, although the representations of visual target object categories were also reliably activated in a top-down manner during the attentional blink interval (indexed by prominent SN component in lag 3 condition), the representation-driven attentional spreading to semantically congruent sounds was also entirely inhibited. Therefore, the present findings demonstrate for the first time that in addition to the previously proposed spatial-based and object-based visual attention operating at the perceptual stage of processing, adequate post-perceptual attentional resource is another vital endogenous prerequisite for the occurrences of both types of attentional spreading from visual to auditory modality.

More specifically, given that the attentional blink has been proposed to reflect most likely an impairment or delay of further consolidation of T2 at the stage of working memory (Chun & Potter, 1995; Dux & Marois, 2009; Dux et al., 2014; Kranczioch et al., 2007; Olivers et al., 2007; Ptito et al., 2008; Raffone et al., 2014; Sergent et al., 2005; Vogel et al., 1998; Zivony & Lamy, 2016), the present inhibitions of cross-modal attentional spreading during the attentional blink might further implicate that not only the audiovisual binding process responsible for the stimulus-driven spread of attention, but also the co-activation of target object category's auditory representation underlying the representation-driven spread of attention, are dependent on the late consolidation processing of visual stimuli in working memory. This interpretation fits with the relatively late

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emergence and the long persistence of both stimulus-driven and representation-driven attentional spreading observed in the present and previous studies (Busse et al., 2005; Donohue et al., 2011; Fiebelkorn, Foxe, & Molholm, 2010; Fiebelkorn, Foxe, Schwartz, et al., 2010; Molholm et al., 2007; Talsma et al., 2007; Zimmer et al., 2010). Indeed, the timings of both types of the cross-modal attentional spreading observed in the previous and present studies suggest that they are more likely to be manifested as the *late* phase of Nd component. The late Nd sub-component has been proposed to reflect an attention-elicited enhancement of higher-order auditory processing than the *early* Nd occurring around the auditory N1 interval (Giard et al., 2000; Hansen & Hillyard, 1980; Näätänen, 1982; Singhal & Fowler, 2004). In particular, previous studies have found that the amplitude of late Nd (but not early Nd) was reduced by introducing an additional and concurrent visual short-term memory task, thereby linking the late Nd with working memory processing (Singhal & Fowler, 2004, 2005). If that is the case, the current findings appear to suggest that what really "spreads" during the cross-modal spread of attention might consist of not merely the spatial-based or object-based visual attention but also the post-perceptual attentional resources necessary for further consolidation processing in visual working memory. Alternatively, it is also possible that the post-perceptual attentional resources are not cross-modally spreading but only serve as a "gate" mechanism for the manifestation of attentional spreading effects. Thus, additional research might be needed to determine whether post-perceptual attentional resources are also embraced as a constituent of the cross-modal spread of attention.

Nevertheless, further visual inspection of Figure 3 and Figure 4c might lead one to speculate that the present effects of attentional blink on stimulus-driven attentional spreading (Figure 3) and representation-driven attentional spreading to auditory-only stimuli (Figure 4c) resulted from ERPs to auditory-only stimuli of nontarget's representations (i.e., A^-) being more negative-going in lag 3 than lag 8 condition. Indeed, a post hoc analysis showed that ERPs to auditory-only stimuli of nontarget's representations were significantly more negative in lag 3 than lag 8 condition within both 200–300 ms interval [$F(1, 37) = 23.33$, $p_{FDR} < .0001$, $\eta^2_p = .39$] and 300–400 ms interval [$F(1, 37) = 24.97$, $p_{FDR} < .0001$, $\eta^2_p = .40$]. However, the way of the interpretation above seems less appropriate because it has been reported that the mismatch negativity (MMN), along with ERP waveforms during 200–400 ms, in response to task-irrelevant sounds presented with visual T2s at lag 3 under attentional blink paradigm was greater (i.e., more negative) when T2s were incorrectly identified (i.e., AB-present) than when T2s were correctly identified (i.e., AB-absent; Haroush et al., 2011). This finding suggests that the *baseline level* of auditory processing of task-irrelevant sounds is inherently different between AB-present and AB-absent trials. The present finding of enlarged negative ERP response to auditory-only stimuli of nontarget's representations in lag 3 than lag 8 condition is consistent with this inference. The present findings indicate that in contrast to the enhanced baseline level of auditory processing during the visual attentional blink (Haroush et al., 2011), both types of further attentional spreading from visual to auditory modality were actually impaired during the attentional blink interval.

In addition to the core findings above, the current study also revealed distinct dependencies on audiovisual semantic congruency for the two types of cross-modal attentional spreading. On the one hand, the stimulus-driven spread of attention in lag 8 condition was found to occur, without magnitude difference, when the visual nontarget T2 was presented synchronously with either a semantically congruent or incongruent sound. On the other hand, the present representation-driven spread of attention in lag 8 condition was found to be evident when the sound was semantically congruent with the visual target T2 (both when the sound was delivered with the visual target and when the sound was delivered alone), but not when the sound was semantically incongruent with the visual target T2. It is noteworthy that similar to Fiebelkorn, Foxe, and Molholm (2010), the incongruent T2 drawings and sounds in the present study were also consistently paired throughout the experiment (e.g., when incongruent, the drawings of dogs were always paired with the beeps of cars; see Method section). Accordingly, had the representation-driven spread of attention been confounded with the effect originating from the temporary associations formed by repeated audiovisual pairings during the experiment, a significant Nd component should have been also evident in response to the incongruent sounds paired with visual targets, but was actually absent, demonstrating that the experiment-specific associations cannot account for the representation-driven spread of attention. Collectively, these findings confirm the existing dual-mechanism model for the cross-modal spread of attention (Fiebelkorn, Foxe, & Molholm, 2010): the stimulus-driven attentional spreading reflects a bottom-up, inherent audiovisual binding mechanism that is independent of audiovisual semantic congruency, whereas the representation-driven attentional spreading demonstrates a top-down cross-modal priming mechanism that is critically dependent on highly learned audiovisual semantic associations.

5 | CONCLUSION

In summary, the current ERP study, taking advantage of the attentional blink paradigm to manipulate the amount of post-perceptual attentional resources, investigated whether the stimulus-driven and representation-driven cross-modal spread of attention would be inhibited when the

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post-perceptual attentional resources are insufficient whereas all visual stimuli are spatially attended and the representations of visual target object categories are activated in a top-down manner. Our data showed that both the stimulus-driven and representation-driven attentional spreading were completely suppressed during the attentional blink interval. These findings provide the first electrophysiological evidence that the occurrences of both stimulus-driven and representation-driven attentional spreading from visual to auditory modality are contingent on the amount of post-perceptual attentional resources responsible for the late consolidation processing of visual stimuli, whereas the early detection of visual stimuli and the top-down activation of the representations of visual targets are not the sole *endogenous* prerequisites for triggering any types of cross-modal attentional spreading. Thus, the present study provides novel insight into the boundary conditions of both types of cross-modal attentional spreading, whereby updating the existing dual-mechanism model (Fiebelkorn, Foxe, & Molholm, 2010) for the cross-modal spread of attention: although the two types of attentional spreading exhibit distinct dependencies on highly learned audiovisual associations as well as on the type of early selective visual attention (spatial-based, object-based), they both depend on the post-perceptual attentional resources.

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CONFLICT OF INTEREST

None.

AUTHOR CONTRIBUTION

Song Zhao: Data curation; Formal analysis; Visualization; Writing-original draft; Writing-review & editing. **Chengzhi Feng:** Data curation; Investigation; Methodology. **Yu Liao:** Formal analysis; Resources; Validation. **Xinyin Huang:** Resources. **Wenfeng Feng:** Conceptualization; Funding acquisition; Supervision; Writing-review & editing.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding authors upon reasonable request.

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SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section.

Figure S1. (a) Illustration of 9 line drawings serving as alternatives for T1, including 3 clothes, 3 cups and 3 flowers. Among the 3 clothes, the first is a long-sleeved blouse, the second is a short-sleeved shirt and the third is a pullover. The 3 cups differ in cup body, handle and saucer. Among the 3 flowers, the first is a lily, the second is a narcissus and the third is a lotus. (b) Separate four alternatives for T2 in target-*dog*, target-*car* and target-*drum* blocks, respectively. Regardless of the block type, T2 could be one of the 9 line drawings shown in this panel (3 dogs, 3 cars and 3 drums) or be blank. The 3 dogs differ mainly in ear, orientation of tail and standing posture. The 3 cars differ mainly in style and orientation of front wheel and in style of radiator grille. The 3 drums differ mainly in bass drum's orientation and the number of snare drums. (c) Illustration of 30 drawings of houses used as distractors for RSVP streams. In each RSVP stream, 15 drawings serving as distractors were sampled randomly from the 30 houses' drawings

Figure S2. ERP waveforms time-locked to T2 onset in *V-only* condition are shown as functions of lag (lag 3, lag 8) and target condition (target, nontarget), which were averaged across six cento-parietal electrodes (CP1, CPz, CP2, P1, Pz, P2). The shaded areas on waveforms depict the time window (400–500 ms) within which the mean amplitude of P3b component was quantified. Scalp topographies are shown for the target minus nontarget difference amplitude within the P3b interval. The white dots on scalp topographies depict the cento-parietal electrodes over which the P3b component was quantified. The measurement parameters for P3b component were chosen *a prior* based on a recent study from our laboratory (Feng et al., 2014). The two-way ANOVA with factors of lag (lag 3, lag 8) and target condition (target, nontarget) conducted on the P3b amplitudes revealed a significant main effect of lag [$F(1, 37) = 105.08$, $p < .0001$, $\eta^2_p = .74$], with P3b amplitude being suppressed overall at lag 3 than lag8 [lag 3: 1.32 ± .38 μV ($M \pm SE$); lag 8: 5.45 ± .44 μV], demonstrating the classic ERP signature of attentional blink (Sergent et al., 2005). The main effect of target condition was significant [$F(1, 37) = 70.38$, $p < .0001$, $\eta^2_p = .66$], with generally larger P3b amplitude in response to target than nontarget categories (target: 5.11 ± .50 μV; nontarget: 1.66 ± .29 μV), indicative of its well-known sensitivity to *post-perceptual* selection (Donchin, 1981). Moreover, the target condition × lag interaction was also significant [$F(1, 37) = 68.26$, $p < .0001$, $\eta^2_p = .65$], which resulted from the P3b target effect (i.e. the target minus nontarget difference) decreasing substantially at lag 3 (1.48 ± .34 μV) relative to lag 8 (5.42 ± .58 μV), indicating again that the post-perceptual attentional selection was impaired during the attentional blink interval (Vogel et al., 1998)

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